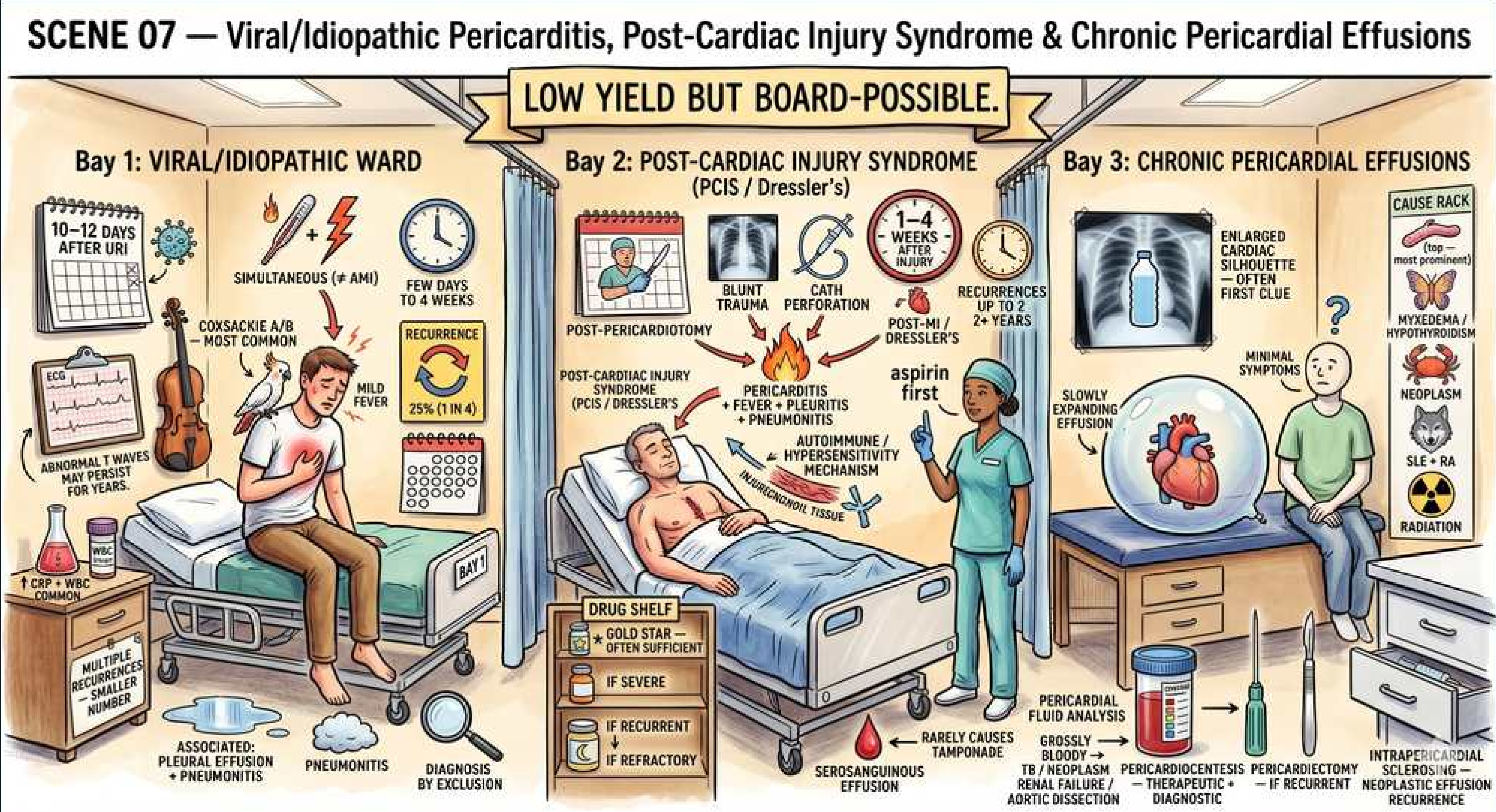
Pericardial Disease — Pericardial Effusion



1. Chronic effusion bay

WHAT YOU SEE
'Bay 3: Chronic Pericardial Effusions' banner

WHAT IT ENCODES
Chronic pericardial effusions accumulate slowly over weeks to months, allowing the parietal pericardium to stretch, so volumes >1 L can be tolerated with minimal symptoms. Normal pericardial fluid is only ~15–50 mL. Symptoms, when present, come from compression of adjacent structures: dysphagia (esophagus), cough/hiccups (phrenic/diaphragm), hoarseness (recurrent laryngeal nerve — Ortner phenomenon).

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2. Slowly expanding

WHAT YOU SEE
Heart inside a balloon labeled 'slowly expanding effusion'

WHAT IT ENCODES
The pericardial pressure-volume relationship is the core concept: slow accumulation shifts the curve rightward (the pericardium remodels and complies), so large chronic effusions stay hemodynamically silent. The same volume added acutely sits on the steep part of the curve and tamponades. Rate of fill > absolute volume in determining tamponade.

BOARD TRAP
WRONG to assume a large effusion equals tamponade and rush to drain a stable patient. A chronic large effusion that built up slowly is often well tolerated; the discriminator for tamponade is HEMODYNAMICS (pulsus paradoxus, RA/RV diastolic collapse on echo, hypotension), not effusion size. Drain for tamponade physiology or diagnosis, not size alone.

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3. Enlarged silhouette

WHAT YOU SEE
CXR with enlarged cardiac silhouette / 'water bottle' shape

WHAT IT ENCODES
An enlarged, globular 'water-bottle' cardiac silhouette on CXR is often the first incidental clue, but the heart looks big only once the effusion exceeds ~200–250 mL. A lateral film may show the 'fat pad'/Oreo-cookie sign (epicardial fat line). ECG correlates: low QRS voltage and, with large effusions, electrical alternans (beat-to-beat QRS amplitude swing from the heart swinging in fluid).

BOARD TRAP
WRONG to attribute an enlarged silhouette to cardiomegaly (dilated cardiomyopathy) and start heart-failure therapy. The discriminator is echocardiography: a globular silhouette with CLEAR lung fields and low-voltage/electrical-alternans ECG points to effusion, not LV dilatation. Always image before treating.

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4. Aspirin first (PCIS)

WHAT YOU SEE
'aspirin first' label in post-cardiac injury bay

WHAT IT ENCODES
In post-MI or post-cardiac-injury effusion/pericarditis, aspirin (650–1000 mg q6–8h, then taper) is the preferred anti-inflammatory because it also serves the patient's antiplatelet indication after MI. Add colchicine 0.5 mg once or twice daily to reduce recurrence. Avoid non-aspirin NSAIDs early post-MI — they impair infarct healing and increase remodeling/rupture risk.

BOARD TRAP
WRONG to give ibuprofen/indomethacin or corticosteroids for post-MI pericarditis. Non-aspirin NSAIDs impair myocardial scar formation (rupture risk) and steroids increase recurrence and impair healing. The discriminator: in the post-MI setting choose ASPIRIN (± colchicine), reserving steroids only for refractory disease.

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5. Echo confirms

WHAT YOU SEE
A specimen cup of drawn pericardial fluid being sent for analysis — the tapped fluid that, with echo, confirms and characterizes the effusion.

WHAT IT ENCODES
Transthoracic echocardiography is the test of choice to confirm, localize, and size an effusion (small <10 mm, moderate 10–20 mm, large >20 mm in diastole) and to assess tamponade physiology (chamber collapse, IVC plethora, respiratory inflow variation). Pericardiocentesis fluid is sent for cell count, protein/LDH, cytology, Gram stain/culture, AFB/adenosine deaminase (TB), and triglycerides (chylopericardium).

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6. Etiology rack

WHAT YOU SEE
'Cause rack': myxedema, neoplasm, SLE, RA, radiation

WHAT IT ENCODES
Chronic/large effusion causes to know: hypothyroidism (myxedema), malignancy (lung, breast, lymphoma, melanoma), autoimmune (SLE, RA, scleroderma), radiation, TB, uremia/dialysis, and idiopathic/viral. Mnemonic for big chronic effusions: 'My MAIN TUbe' — Myxedema, Malignancy, Autoimmune, Idiopathic, Nephrons (uremia), TB, Uremia.

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7. Hypothyroid cause

WHAT YOU SEE
Myxedema / hypothyroidism listed on cause rack

WHAT IT ENCODES
Myxedema produces large, slow, cholesterol-rich effusions that are classically asymptomatic and rarely tamponade. The fluid is high in protein/cholesterol. It resolves with thyroid hormone replacement rather than drainage. Check TSH in any unexplained large chronic effusion.

BOARD TRAP
WRONG to proceed straight to pericardiocentesis for a large effusion before checking thyroid function. The discriminator: a huge but hemodynamically stable effusion with high cholesterol content and an elevated TSH is myxedema — treat with levothyroxine; the effusion regresses, drainage is usually unnecessary.

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8. Malignant cause

WHAT YOU SEE
Neoplasm icon on cause rack

WHAT IT ENCODES
Malignancy is a leading cause of large and recurrent BLOODY effusions; lung and breast cancer (direct/contiguous spread) and lymphoma/leukemia are most common, plus melanoma. Cytology is positive in ~50–80%; pericardial fluid CEA may help. Recurrent malignant effusions are managed with a pericardial window or balloon pericardiotomy ± intrapericardial sclerosing/chemo agents.

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9. Grossly bloody → malignancy

WHAT YOU SEE

'Grossly bloody' fluid note near specimen

WHAT IT ENCODES

Grossly bloody (hemorrhagic) pericardial fluid narrows the differential to malignancy, TB, recent cardiac procedure/trauma, aortic dissection rupturing into the pericardium, or post-MI free-wall rupture. Send cytology and AFB/ADA. An effusion that is frankly bloody plus hemodynamic compromise after a procedure should raise dissection or rupture immediately.

BOARD TRAP

WRONG to write off bloody fluid as a 'traumatic tap.' True hemorrhagic effusion (hematocrit approaching peripheral blood, doesn't clot if chronic) signals malignancy, TB, dissection, or rupture. The discriminator: send cytology/ADA and consider CT angiography for dissection — do not dismiss it as artifact.

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10. Pericardiocentesis if recurrent

WHAT YOU SEE

'Pericardiocentesis — therapeutic if recurrent' note

WHAT IT ENCODES

Pericardiocentesis is both diagnostic (fluid analysis) and therapeutic (relieves tamponade). For recurrent neoplastic effusions, definitive options are a surgical pericardial window (pericardio-pleural/peritoneal drainage), percutaneous balloon pericardiotomy, or intrapericardial sclerotherapy/instillation. Echo or fluoroscopic guidance reduces complication risk.

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11. Rarely seroug./serous

WHAT YOU SEE

A label reading 'serosanguinous effusion / rarely causes tamponade' beside the fluid — the straw-pink fluid color keys you to the etiology ladder.

WHAT IT ENCODES

Fluid character helps narrow etiology: serous (transudative — heart failure, myxedema, nephrotic), serosanguinous (TB, malignancy, post-injury), grossly bloody (malignancy/TB/dissection/trauma), purulent (bacterial — emergency), milky (chylopericardium — high triglycerides). Apply Light's-type criteria; exudates point to inflammatory/infectious/malignant causes.

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12. Viral/idiopathic ward

WHAT YOU SEE

'Bay 1: Viral/Idiopathic Ward' with violin

WHAT IT ENCODES

Viral/idiopathic pericarditis is the most common overall cause of acute pericarditis and effusion in developed countries, typically 1–3 weeks after a viral URI or GI illness (Coxsackie B, echovirus, adenovirus). It is usually self-limited; treat with NSAIDs + colchicine. Most associated effusions are small and resolve.

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13. Effusion with pneumonitis

WHAT YOU SEE

'Associated: pleural effusion + pneumonitis' note

WHAT IT ENCODES

Viral pericarditis may be part of a broader serositis with concurrent pleural effusion and pneumonitis (myopericarditis if the myocardium is involved — watch troponin). Pleuritic, positional chest pain (worse supine, better leaning forward) and a three-component friction rub are the bedside hallmarks.

BOARD TRAP

WRONG to call elevated troponin in this setting an acute coronary syndrome and rush to catheterization. The discriminator: diffuse concave ST elevation with PR depression, a pericardial rub, positional pleuritic pain, and normal coronaries on imaging point to myopericarditis — manage as pericarditis, not ACS.

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14. Diagnosis of exclusion

WHAT YOU SEE

'Diagnosis by exclusion' note in Bay 1

WHAT IT ENCODES

Idiopathic pericarditis/effusion is a diagnosis of exclusion made after a focused workup is unrevealing. Reasonable baseline workup: ECG, echo, CXR, CBC, troponin, ESR/CRP, renal function, and TSH; pursue cytology/ADA/cultures and autoimmune serologies only if features suggest a specific cause. Most do not need extensive testing.

BOARD TRAP

WRONG to launch an exhaustive search for malignancy/autoimmune disease in a young patient with a typical small self-limited effusion. The discriminator: pursue aggressive workup (cytology, ADA, biopsy) only when there are high-risk features — large/recurrent effusion, fever >38°C, subacute course, immunosuppression, anticoagulation, or failure of NSAIDs after a week.
